

American International University–Bangladesh (AIUB)
Department of Computer Science, Faculty of Science and Technology
CN Lecture 04: Multiple Access Techniques II (Data Link Layer II)
OBE Based Questions and Answers

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1. A university has implemented a shared communication channel where multiple computers transmit data without checking whether the channel is busy. Analyze the problems that may occur and explain how the ALOHA protocol handles them.

Answer:

In ALOHA, a station transmits whenever it has data without sensing the channel.

This may result in:

- Frequent frame collisions.
- Loss of transmitted data.
- Increased retransmissions.
- Lower channel efficiency.

ALOHA handles collisions by waiting for an acknowledgment (ACK). If no ACK is received within the timeout period, the sender assumes a collision has occurred, waits for a random backoff time, and retransmits the frame.

2. An engineer wants to improve the performance of Pure ALOHA without changing its basic communication principle. Recommend an appropriate protocol and justify your recommendation.

Answer:

The appropriate protocol is **Slotted ALOHA**.

In Slotted ALOHA:

- Time is divided into equal slots.
- A station can transmit only at the beginning of a slot.
- Random transmissions are eliminated.
- The vulnerable time is reduced from $2T_{fr}$ to T_{fr} .

Therefore, Slotted ALOHA significantly reduces collisions and improves channel efficiency.

3. Compare Pure ALOHA and Slotted ALOHA with respect to transmission strategy, vulnerable time, and network efficiency.

Answer:

Feature	Pure ALOHA	Slotted ALOHA
Transmission Time	Anytime	Beginning of a slot
Synchronization	Not required	Required
Vulnerable Time	$2T_{fr}$	T_{fr}
Collision Probability	Higher	Lower
Efficiency	Lower	Higher

Slotted ALOHA is more efficient because it reduces the vulnerable period and minimizes collisions.

4. A company wants to minimize frame collisions in a shared Ethernet network. Explain how CSMA improves upon ALOHA.

Answer:

CSMA follows the principle of **Carrier Sense**.

Before transmitting, a station checks whether the communication channel is idle.

- If the channel is idle, transmission begins.
- If the channel is busy, the station waits according to the selected persistence strategy.

By sensing the channel before transmission, CSMA significantly reduces the probability of collisions compared to ALOHA.

5. **Although CSMA senses the channel before transmission, collisions can still occur. Analyze the reason behind this limitation.**

Answer:

Collisions still occur because of **propagation delay**.

When one station starts transmitting, the signal requires time to reach other stations.

During this propagation interval, another station may still sense the channel as idle and begin transmission.

As a result, both transmissions collide.

Therefore, propagation delay is the primary cause of collisions in CSMA networks.

6. **A network administrator is choosing among 1-persistent, Nonpersistent, and p-persistent CSMA. Compare these methods and recommend the most balanced approach.**

Answer:

Feature	1-Persistent	Nonpersistent	p-Persistent
Transmit Immediately	Yes	Yes (if idle)	With probability p
Collision Chance	High	Low	Moderate
Efficiency	Moderate	Lower	Higher
Waiting Strategy	Continuous sensing	Random waiting	Probability-based

Among the three, p-persistent CSMA provides the best balance between collision reduction and efficient channel utilization.

7. **A busy office network experiences frequent simultaneous transmissions whenever the channel becomes idle. Which CSMA persistence strategy should be avoided? Explain your answer.**

Answer:

The **1-persistent CSMA** strategy should be avoided.

In this method:

- Every station continuously senses the channel.
- As soon as the channel becomes idle, all waiting stations transmit immediately.
- Multiple simultaneous transmissions increase collision probability.

Therefore, 1-persistent CSMA performs poorly under heavy network traffic.

8. **An Ethernet network requires immediate detection of collisions during transmission. Explain how CSMA/CD achieves this objective.**

Answer:

CSMA/CD operates as follows:

- Sense the communication channel.
- Transmit if the channel is idle.
- Continue monitoring the channel during transmission.
- Detect collisions immediately.
- Stop transmitting when a collision occurs.
- Send a Jam Signal.
- Wait for a random backoff period.
- Retransmit the frame.

This approach minimizes bandwidth wastage caused by corrupted transmissions.

9. **Explain the purpose of the Jam Signal in CSMA/CD.**

Answer:

A Jam Signal is transmitted immediately after a collision is detected.

Its purposes are:

- Inform all stations that a collision has occurred.
- Ensure all transmitting stations stop transmission.
- Prevent corrupted frames from continuing.
- Initiate the random backoff procedure.

Thus, the Jam Signal improves collision recovery in wired Ethernet networks.

10. **A wireless LAN cannot use CSMA/CD. Analyze the reasons and explain why CSMA/CA is preferred instead.**

Answer:

Wireless devices cannot reliably detect collisions because:

- Signal fading.
- Hidden terminal problem.
- Large transmission distance.
- Half-duplex communication.
- Battery limitations.

Therefore, wireless networks avoid collisions rather than detecting them.

CSMA/CA is specifically designed for wireless communication.

11. **Explain how CSMA/CA prevents collisions before data transmission in a wireless network.**

Answer:

CSMA/CA uses three collision avoidance mechanisms.

- (a) **Interframe Space (IFS):** The sender waits for a fixed period after the channel becomes idle.
- (b) **Contention Window:** Each station selects a random backoff value before transmission.
- (c) **Acknowledgment (ACK):** The receiver confirms successful frame reception. If no ACK is received, retransmission occurs.

These mechanisms greatly reduce the probability of collisions.

12. **Evaluate the suitability of ALOHA, CSMA/CD, and CSMA/CA for different networking environments.**

Answer:

Protocol	Suitable Environment
ALOHA	Simple shared communication systems with low traffic
CSMA/CD	Wired Ethernet LANs where collisions can be detected
CSMA/CA	Wireless LANs where collision detection is impractical

ALOHA is simple but inefficient.

CSMA/CD is highly effective for wired networks.

CSMA/CA is the preferred solution for modern wireless communication because it prevents collisions before they occur.